

Spherical Astronomy, Coordinates, and Time

Problem Set — Solutions

fizicyst

Section A — Objective

O1. Answer: (b). $LST = H + \alpha = 2^{\text{h}}15^{\text{m}} + 5^{\text{h}}30^{\text{m}} = 7^{\text{h}}45^{\text{m}}$.

O2. Answer: (b). At the pole the NCP sits at the zenith, so the celestial equator — 90° from it — runs exactly along the horizon.

O3. Answer: (a). $h_u = 90^\circ - |\phi - \delta| = 90^\circ - |40^\circ - 70^\circ| = 60^\circ$.

O4. Answer: (d) is false. Any spherical triangle with nonzero area has strictly positive angular excess, since $A_{\text{area}} = ER^2$. Sphere size rescales the area but never makes E vanish. (a), (b), (c) are all correct.

O5. Answer: (a). $\Theta = \cos^{-1}\left(\frac{R_\oplus}{R_\oplus + x}\right) = \cos^{-1}\left(\frac{6371}{6371.045}\right) \approx 3.76 \times 10^{-3} \text{ rad} \approx 0.215^\circ \approx 13'$.

Section B — Subjective

S1.

(i) At $H = 0$ the altitude formula collapses to $\sin h = \cos(\phi - \delta)$, so

$$h_u = 90^\circ - |\phi - \delta| = 90^\circ - |20^\circ - 35^\circ| = 75^\circ.$$

(ii) Setting $h = 0$: $\cos H = -\tan \phi \tan \delta = -\tan 20^\circ \tan 35^\circ = -(0.3640)(0.7002) \approx -0.2549$, so

$$H \approx 104.76^\circ = 6.984^{\text{h}} \approx 6^{\text{h}}59^{\text{m}}.$$

Checks: $|\phi + \delta| = 55^\circ < 90^\circ$ (not circumpolar) and $|\phi - \delta| = 15^\circ < 90^\circ$ (not a non-riser), so the star rises and sets normally — consistent with $|\cos H| < 1$ having a real solution at all.

(iii) At rising, $H = -6.984^{\text{h}}$, and $\alpha = 6^{\text{h}}40^{\text{m}} = 6.667^{\text{h}}$:

$$LST = H + \alpha = -6.984 + 6.667 = -0.317^{\text{h}} \equiv 23.683^{\text{h}} \approx 23^{\text{h}}41^{\text{m}}.$$

(The negative result wraps around by adding 24^{h} .)

S2.

- (i) At $H = 0^\circ$, $\cos H = 1$:

$$\sin h_u = \sin \phi \sin \delta + \cos \phi \cos \delta = \cos(\phi - \delta).$$

At $H = 180^\circ$, $\cos H = -1$:

$$\sin h_l = \sin \phi \sin \delta - \cos \phi \cos \delta = -\cos(\phi + \delta).$$

- (ii) Since $\cos(\phi - \delta) = \sin(90^\circ - |\phi - \delta|)$, the first gives $h_u = 90^\circ - |\phi - \delta|$. Since $-\cos(\phi + \delta) = \sin((\phi + \delta) - 90^\circ)$, the second gives $h_l = |\phi + \delta| - 90^\circ$ (the absolute value is harmless given $\phi, \delta > 0$ and the star circumpolar, so $\phi + \delta > 90^\circ$).
- (iii) Both relations are linear in ϕ and δ . Taking the case $\delta > \phi$ (the other case is identical up to signs):

$$h_u = 90^\circ - \phi + \delta, \quad h_l = \phi + \delta - 90^\circ.$$

As a system in (ϕ, δ) the coefficient matrix is $\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$, whose determinant is $-2 \neq 0$ (the other case gives $+2$). A nonzero determinant guarantees the system is invertible, so (ϕ, δ) is uniquely determined by (h_u, h_l) — no choice of measured altitudes can ever give an ambiguous or degenerate answer. Inverting recovers exactly Section 6.2's results, $\delta = (h_u + h_l)/2$ and $\phi = (h_l - h_u + 180^\circ)/2$.

S3.

- (i) $\sin \delta_\odot = \sin \epsilon \sin \lambda_\odot = \sin 23.5^\circ \sin 60^\circ = (0.3988)(0.8660) \approx 0.3453$, so $\delta_\odot \approx 20.2^\circ\text{N}$.
- (ii) $\tan \alpha_\odot = \cos \epsilon \tan \lambda_\odot = (0.9171)(1.7321) \approx 1.5884$, giving $\alpha_\odot \approx 57.8^\circ$. Since $\lambda_\odot = 60^\circ$ is in the first quadrant, so is $\alpha_\odot$, and no 180° correction is needed:

$$\alpha_\odot \approx 57.8^\circ = 3.853^{\text{h}} \approx 3^{\text{h}}51^{\text{m}}.$$

- (iii) $\cos H = -\tan \phi \tan \delta_\odot = -\tan 35^\circ \tan 20.2^\circ = -(0.7002)(0.3679) \approx -0.2576$, so $H \approx 104.93^\circ \approx 6.995^{\text{h}}$. Daylight runs from $-H$ to $+H$, so

$$\text{day length} = 2H \approx 13.99 \approx 14.0 \text{ hours.}$$

(Sensible for a northern-hemisphere location about two-thirds of the way from equinox to summer solstice.)